

## YEAGER, WV, USA

WMO: 724140

Lat: 38.379N

Lon: 81.590W

Elev: 277

StdP: 98.04

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13866

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.4      | -9.6 | -18.4                         | 0.8 | -9.6 | -15.6 | 1.0 | -7.7 | 9.2                   | 5.2  | 8.3 | 5.9  | 1.6                   | 210  | 0.404 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.5                   | 32.8            | 22.7 | 31.6 | 22.4 | 30.4 | 22.0 | 24.8                | 29.8 | 24.0 | 28.8 | 23.4 | 27.9 | 2.7                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.3                            | 18.7 | 26.9 | 22.6 | 17.9 | 26.0 | 22.0 | 17.3 | 25.4 | 76.3          | 30.1 | 73.4 | 29.0 | 70.8 | 27.9 | 28.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 8.0               | 6.8  | 5.6 | DB                         | -16.6 | 34.7               | 3.7 | 1.9                                                | -19.3 | 36.1       | -21.4 | 37.2       | -23.5 | 38.3       | -26.2 | 39.7 |  |
|                   |      |     | WB                         | -17.0 | 26.1               | 3.6 | 1.0                                                | -19.6 | 26.8       | -21.7 | 27.4       | -23.7 | 27.9       | -26.4 | 28.6 |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 13.3   | 1.4   | 3.3   | 7.7   | 13.9  | 18.1  | 22.3  | 23.9  | 23.5  | 20.1  | 13.8  | 7.9   | 3.7  |
|                                                                               | DBStd       | 9.30   | 6.76  | 6.16  | 6.04  | 5.20  | 4.19  | 2.91  | 2.44  | 2.43  | 3.60  | 4.72  | 5.13  | 5.64 |
|                                                                               | HDD10.0     | 944    | 279   | 198   | 118   | 21    | 2     | 0     | 0     | 0     | 0     | 17    | 100   | 208  |
|                                                                               | HDD18.3     | 2436   | 526   | 420   | 334   | 149   | 59    | 5     | 0     | 1     | 23    | 153   | 313   | 453  |
|                                                                               | CDD10.0     | 2165   | 12    | 12    | 46    | 138   | 252   | 368   | 432   | 418   | 302   | 136   | 37    | 13   |
|                                                                               | CDD18.3     | 616    | 0     | 0     | 3     | 16    | 51    | 124   | 173   | 161   | 75    | 13    | 1     | 0    |
|                                                                               | CDH23.3     | 5308   | 1     | 2     | 34    | 218   | 487   | 1049  | 1446  | 1282  | 638   | 142   | 9     | 0    |
|                                                                               | CDH26.7     | 1785   | 0     | 0     | 4     | 48    | 136   | 352   | 536   | 460   | 219   | 29    | 1     | 0    |
| Wind                                                                          | WSAvg       | 1.9    | 2.4   | 2.4   | 2.4   | 2.4   | 1.8   | 1.6   | 1.4   | 1.2   | 1.2   | 1.5   | 1.8   | 2.1  |
| Precipitation                                                                 | PrecAvg     | 1116   | 79    | 78    | 100   | 86    | 106   | 104   | 138   | 100   | 91    | 71    | 81    | 85   |
|                                                                               | PrecMax     | 1705   | 165   | 197   | 211   | 160   | 222   | 271   | 361   | 262   | 295   | 155   | 232   | 204  |
|                                                                               | PrecMin     | 742    | 25    | 13    | 33    | 13    | 21    | 16    | 50    | 8     | 27    | 3     | 15    | 9    |
|                                                                               | PrecStd     | 192    | 36    | 38    | 41    | 30    | 53    | 51    | 61    | 47    | 55    | 37    | 38    | 41   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 20.7  | 22.6  | 26.9  | 30.3  | 31.6  | 33.4  | 34.5  | 34.1  | 33.5  | 29.7  | 24.7  | 21.0 |
|                                                                               |             | MCWB   | 14.7  | 15.0  | 16.3  | 17.3  | 20.7  | 22.5  | 24.0  | 23.2  | 21.1  | 20.0  | 16.2  | 14.9 |
|                                                                               | 2%          | DB     | 17.6  | 18.8  | 23.8  | 28.1  | 29.7  | 31.6  | 32.7  | 32.3  | 31.4  | 26.9  | 21.9  | 17.9 |
|                                                                               |             | MCWB   | 13.2  | 12.5  | 14.6  | 16.6  | 20.3  | 22.4  | 23.7  | 22.9  | 21.3  | 18.9  | 14.6  | 13.1 |
|                                                                               | 5%          | DB     | 15.0  | 16.1  | 20.8  | 26.1  | 28.3  | 30.3  | 31.2  | 30.9  | 29.3  | 24.7  | 19.5  | 15.6 |
|                                                                               |             | MCWB   | 11.7  | 10.9  | 12.7  | 15.7  | 19.8  | 22.0  | 23.2  | 22.6  | 20.8  | 17.2  | 13.2  | 11.7 |
|                                                                               | 10%         | DB     | 12.0  | 13.2  | 17.7  | 23.6  | 26.3  | 28.9  | 29.8  | 29.4  | 27.2  | 22.2  | 17.0  | 13.1 |
|                                                                               |             | MCWB   | 8.8   | 9.3   | 11.5  | 14.7  | 19.2  | 21.4  | 22.7  | 22.3  | 20.0  | 16.5  | 12.0  | 9.8  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.8  | 16.2  | 17.3  | 19.9  | 23.0  | 25.0  | 26.0  | 25.6  | 24.1  | 22.1  | 17.8  | 15.8 |
|                                                                               |             | MCDB   | 19.3  | 20.9  | 25.2  | 27.0  | 28.5  | 30.2  | 31.8  | 30.4  | 29.2  | 26.1  | 22.7  | 19.4 |
|                                                                               | 2%          | WB     | 14.0  | 14.0  | 15.6  | 18.2  | 21.8  | 23.9  | 24.9  | 24.6  | 22.9  | 20.4  | 15.9  | 14.2 |
|                                                                               |             | MCDB   | 16.8  | 17.2  | 21.3  | 24.5  | 27.3  | 29.0  | 30.3  | 29.3  | 27.2  | 24.7  | 20.0  | 17.1 |
|                                                                               | 5%          | WB     | 12.1  | 11.8  | 14.4  | 17.1  | 20.9  | 23.0  | 24.1  | 23.9  | 21.9  | 18.8  | 14.3  | 12.4 |
|                                                                               |             | MCDB   | 15.0  | 15.1  | 19.2  | 23.0  | 26.0  | 28.1  | 28.9  | 28.5  | 26.2  | 22.5  | 18.0  | 14.8 |
|                                                                               | 10%         | WB     | 9.1   | 9.5   | 12.7  | 15.9  | 20.0  | 22.4  | 23.5  | 23.1  | 21.1  | 17.4  | 12.7  | 10.2 |
|                                                                               |             | MCDB   | 11.8  | 12.8  | 16.9  | 21.5  | 24.7  | 27.1  | 27.9  | 27.3  | 25.3  | 21.0  | 16.4  | 12.7 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.1   | 9.9   | 11.3  | 13.0  | 11.9  | 11.1  | 10.5  | 10.8  | 11.7  | 12.1  | 11.3  | 9.2  |
|                                                                               |             | MCDBR  | 11.9  | 14.8  | 16.2  | 16.4  | 14.6  | 13.2  | 12.4  | 12.9  | 14.6  | 15.2  | 15.3  | 12.8 |
|                                                                               |             | MCWBR  | 8.2   | 10.0  | 8.8   | 7.3   | 6.3   | 5.3   | 4.7   | 4.7   | 5.5   | 7.3   | 8.9   | 8.8  |
|                                                                               | 5% WB       | MCDBR  | 10.7  | 13.0  | 13.0  | 12.7  | 11.8  | 10.9  | 10.4  | 10.6  | 11.5  | 12.0  | 12.1  | 11.0 |
|                                                                               |             | MCWBR  | 8.0   | 10.0  | 8.0   | 6.6   | 5.6   | 4.9   | 4.5   | 4.5   | 5.0   | 6.5   | 8.1   | 8.4  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.307  | 0.322 | 0.347 | 0.389 | 0.436 | 0.460 | 0.488 | 0.491 | 0.401 | 0.360 | 0.325 | 0.300 |      |
|                                                                               | taud        | 2.495  | 2.442 | 2.429 | 2.323 | 2.194 | 2.172 | 2.114 | 2.079 | 2.324 | 2.439 | 2.513 | 2.540 |      |
|                                                                               | Ebn at Noon | 876    | 906   | 914   | 889   | 849   | 826   | 799   | 785   | 844   | 849   | 847   | 861   |      |
|                                                                               | Edh at Noon | 79     | 94    | 106   | 125   | 145   | 148   | 156   | 158   | 116   | 94    | 77    | 71    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1.79   | 2.48  | 3.61  | 4.81  | 5.46  | 6.03  | 5.67  | 5.33  | 4.46  | 3.21  | 2.28  | 1.58  |      |
|                                                                               | RadStd      | 0.20   | 0.35  | 0.28  | 0.39  | 0.45  | 0.35  | 0.32  | 0.28  | 0.44  | 0.40  | 0.28  | 0.19  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | +0.50 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +150    | +92     |
| (47) | Regional (26 neighbors) | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +132    | +66     |

Nomenclature: See separate page